

Scoring Guide

2011 Mathematics · Grade 4 · Block M8 · Item 2011-4M8 · Constructed response

An amusement park has games, rides, and shows. The total number of games, rides, and shows is 70. There are 34 rides. There are two times as many games as shows. How many games are there? How many shows are there? Use numbers, words, or drawings to show how you got your answer.

Score level	Description
Extended	24 games and 12 shows, with correct explanation or work.
Satisfactory	Subtraction error but games and shows in the correct 2 : 1 ratio; OR 12 games and 24 shows with work; OR 24 games and 12 shows with no work.
Partial	Finds 36 and a 2 : 1 ratio (but not 24 : 12) with a sum under 36; OR 36 and 18; OR 72 and 36; OR a process that shows understanding but not the correct ratio.
Minimal	Finds 36 by subtracting (or adding up to 70); OR games + shows = 36; OR a 2 : 1 ratio but nothing else correct.
Incorrect	Incorrect response.

Source: NAEP Questions Tool, released item 2011-4M8 #19. Public domain. Holistic; built for this task only.

Your table's rubric. Score all four work samples with this rubric and nothing else. For each, record the score and the reason it earned that score. We pool every table's scores in the debrief.

Maryland Math Communication Rubric

Source: Maryland State Department of Education, *Sample activities, student responses and Maryland teachers' comments on a sample task: Mathematics Grade 8*, February 1991.

Subjects:	Mathematics	# of scales	1
Grade(s)	8	Scale length	5

Holistic Scale

- 4 Uses mathematical language (terms, symbols, representations) that is highly effective, accurate, and thorough, to describe operations, concepts, and processes.
- 3 Uses mathematical language that is partially effective, accurate, and thorough.
- 2 Uses mathematical language that is minimally effective and accurate.
- 1 An incorrect response – an attempt is made.
- 0 Off task, off topic, illegible, blank, or insufficient to score.

Your table's rubric. Score all four work samples with this rubric and nothing else. For each, record the score and the reason it earned that score. We pool every table's scores in the debrief.

TEMPLE INDEPENDENT SCHOOL DISTRICT · TEMPLE, TEXAS

Mathematics Rubric

Holistic scale · one score per response · 0–3

3	Response is exemplary, detailed and clear.
2	Response is generally correct.
1	Response is partially correct, but lacks clarity.
0	No response, or response is incorrect.

Source: Temple Independent School District, Temple, Texas. Subjects: Mathematics · Grade(s): not specified · 1 scale · scale length 4.

Your table's rubric. Score all four work samples with this rubric and nothing else. For each, record the score and the reason it earned that score. We pool every table's scores in the debrief.

Fig. 3 A general rubric, which is not task-specific, can be used to assess a variety of tasks.

ANALYTIC SCORING SCALE

Understanding the problem

- 6 Complete understanding of the problem.
- 3 Part of the problem misunderstood or misinterpreted.
- 0 Complete misunderstanding of the problem.

Planning a solution

- 6 Plan could have led to a correct solution if implemented properly.
- 3 Partially correct plan based on correct interpretation of part of the problem.
- 0 No attempt, or totally inappropriate plan.

Getting an answer

- 3 Correct answer and correct label for the answer.
- 2 Incorrect answer that follows logically from an inappropriate plan.
- 1 Copying error, computational error, or partial answer (problem with multiple answers).
- 0 No answer, or wrong answer based on an inappropriate plan.

Source: Adapted from Charles, Lester, and O'Daffer 1987, as shown in McGatha, M. B., & Darcy, P. (2010). Rubrics at play. *Mathematics Teaching in the Middle School*, 15(6), 328–336, Figure 3. Analytic; a separate score for each part of problem solving; 15 points possible.

Your table's rubric. Score all four work samples with this rubric and nothing else. For each, record the three scores, the total, and the reason each earned its score. We pool every table's scores in the debrief.